

RESEARCH STATEMENT

Towards a p -adic spectral geometry

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OVERVIEW

Throughout my mathematical research, I have always been drawn to *arithmetic phenomena* first encountered analytically – values of automorphic functions, meromorphic modular forms, Laplacian spectral patterns – that seem to have a deep *geometric nature* involving moduli spaces and motives.

Consider the Riemann surface $\Gamma \backslash \mathbb{H}$ obtained as a quotient of the upper-half complex plane $\mathbb{H} = \{z = x + iy : y > 0\}$ by Möbius transformations under a Fuchsian group $\Gamma \subseteq \mathrm{PSL}_2(\mathbb{R})$. The Laplace-Beltrami operator Δ is explicitly given by

$$\Delta = -y^2(\partial_x^2 + \partial_y^2). \quad (1)$$

This operator classically governs diffusion and wave propagation on a negatively curved space. On a hyperbolic surface, this captures the dynamics, the geometry and, when the surface is arithmetic, the number theory of the space. Its eigenfunctions $\Delta\varphi = \lambda\varphi$, called *Maaß forms*, are hyperbolic waves (Figure 1) and their nodal domains (see Figure 3) represent the geometry of sign changes of these waves.

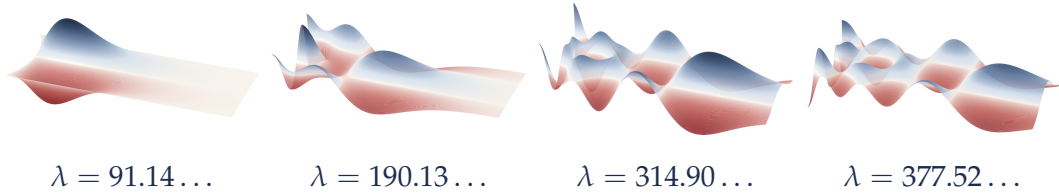


Figure 1: Four Maaß cusp forms for $\mathrm{SL}_2(\mathbb{Z})$, with Laplace eigenvalues λ over the fundamental domain $\{|\mathrm{Re} z| \leq \frac{1}{2}, |z| \geq 1\}$, with the cusp at the right.

In the arithmetic case, for example the *modular curve* $Y(1) = \mathrm{SL}_2(\mathbb{Z}) \backslash \mathbb{H}$, these analytic objects unexpectedly acquire deep arithmetic meaning central in modern number theory:

1. Hecke-Maaß eigenforms conjecturally encode Galois representations and L -functions,
2. special values of automorphic Green kernels connect L -functions with heights and intersections of special algebraic cycles.

The first is an incarnation of the *Langlands philosophy* while the second lies behind the Gross–Zagier–Kolyvagin approach to the rank-one *Birch and Swinnerton-Dyer conjecture*.

My research asks what survives from this picture in p -adic geometry. There is no single p -adic Laplacian playing all of these roles. Together, these projects form a programme toward a p -adic spectral geometry of modular and Shimura curves.

- **Project 1.1** studies *p -adic Green functions* on the Drinfeld upper half-plane, develops explicit Nekovář *p -adic heights* of CM cycles on Shimura curves and proves a *p -adic Gross–Kohnen–Zagier formula*,
- **Project 1.2** studies *extension of filtered φ -modules* arising from modular curves and gives an explicit comparison between Frobenius on cohomology with U_p -operators on meromorphic modular forms. These extensions appear in the context of p -adic heights.
- **Project 1.3** studies the *spectral analogies* between the Laplacian operator on Maaß forms and the

U_p operator on *overconvergent modular forms*. It focuses on the slope distribution and on the zero loci.

1 PHD WORK:

THREE p -ADIC APPROACHES TO ANALYTIC PHENOMENA

1.1 p -adic heights on Shimura curves and higher p -adic Green functions [BH26]

[Joint preprint with [Hazem Hassan](#), available [here](#) (49 pages)]

The value of higher Green functions at CM points has been conjectured in [GKZ87] to be equal to the *logarithm of an algebraic number*. This has sparked research in several directions and has now recently been fully resolved in many instances by fairly different methods. The nature of the object has indeed attracted researchers both from a more analytic side [BEY21], [Via11] and [Li23] via the theory of Borchers lifts and analytic Maaß forms and from a geometric one [Zha97], [Mel08] and [BF25] using a more motivic approach. This latter approach gave us a natural template for a p -adic construction: if higher Green's functions are periods of extensions in the archimedean setting, their p -adic analogues should be p -adic heights of the corresponding extensions of Galois representations, in the sense of Nekovář [Nek93] and Coleman–Gross.

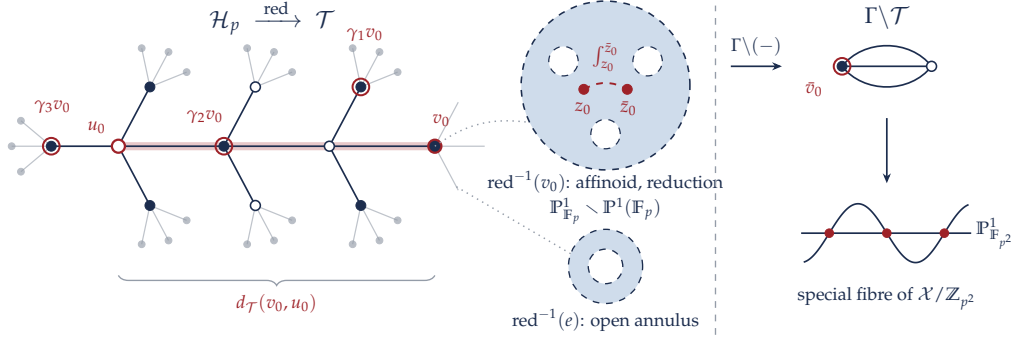


Figure 2: **The Čerednik–Drinfeld uniformisation at a prime p .** *Left.* Drinfeld’s upper half plane $\mathcal{H}_p = \mathbb{P}_{\mathbb{C}_p}^1 \setminus \mathbb{P}^1(\mathbb{Q}_p)$ retracts onto the $(p+1)$ -regular Bruhat–Tits tree $\mathcal{T}$ of $\mathrm{PGL}_2(\mathbb{Q}_p)$. The fibre of red over a vertex is an affinoid with canonical reduction $\mathbb{P}_{\mathbb{F}_p}^1 \setminus \mathbb{P}^1(\mathbb{F}_p)$ while over an open edge is an open annulus. *Right.* $X_{\mathbb{Q}_{p^2}} \cong \Gamma \backslash \mathcal{H}_p$. The finite graph $\Gamma \backslash \mathcal{T}$ is the dual graph of the special fibre, whose components are $\mathbb{P}_{\mathbb{F}_{p^2}}^1$ meeting at ordinary double points.

In joint work with Hazem Hassan, I have developed a p -adic analogue of the higher Green functions $G_k(z, w)$, and realized them as explicit local p -adic heights of CM cycles. Working on Shimura curves uniformized p -adically via Čerednik–Drinfeld (Figure 2) we constructed extensions supported at CM points, where p is inert, producing *biextensions of semistable Galois representations* to which Nekovář’s general local height pairing formalism applies.

Local height formula

In the paper we have extended Besser’s [Bes02] [Bes22] and Besser–Zerbes’s [BZ23] p -adic height approach from trivial to arbitrary symmetric-power coefficients, and we adapt Brown–Fonseca’s generating-series computation of the archimedean primitive to the p -adic, rigid-analytic setting. The resulting formula expresses the p -adic height first as the Coleman integral of a Poincaré series Ψ_k and then as a sum over the uniformizing arithmetic group Γ

$$G_k^{(p)}(z_0, w_0) = \int_{z_0}^{z_0} \Psi_k(z, w_0) dz = \sum_{\gamma \in \Gamma} Q_m(1 - C(\gamma z_0, w_0)). \quad (2)$$

with Q_m the Legendre function of the second kind and C the cross ratio.

Analytic realization and algebraicity

This approach gives a blueprint to prove the p -adic log algebraicity of the p -adic higher Green functions following [BF25]. We can act by the Hecke algebra on the global cohomology to make the biextension a simple extension of Tate motives, i.e. a *Kummer motive*.

Theorem 1 ([BH26] p -adic Gross – Kohnen – Zagier conjecture). *Fix the Shimura curve X and the positive even integer k . Consider a non-zero finite linear combination of Hecke operators T which kills V . Let K_1, K_2 be imaginary quadratic fields with coprime discriminants and H_1, H_2 their Hilbert class field. Then there is an element $\alpha_T \in H = H_1 H_2$ for which the sum over all Heegner points z_0, w_0 associated to K_1, K_2 of the evaluation of the higher Green's function is equal to*

$$\mathrm{Tr}_{H_v/\mathbb{Q}_p} \left(\sum_{z_0, w_0} G_k^{(p)}(z_0, T w_0) \right) = C_{k, K_1, K_2} \log_p(N_{H/\mathbb{Q}}(\alpha_T)) \quad (3)$$

for an explicit constant C_{k, K_1, K_2} .

1.2 A p -adic cohomological approach to congruences of meromorphic modular forms

[Single-author preprint available on [arXiv](#) (12 pages)]

Arithmetic phenomenon

In order to explain the content of the paper and the main result we can start with the following phenomenon first observed and conjectured by Pengchang Zhang (MPIM, IHES) in [Zha25]. Consider the following elliptic curve and the meromorphic modular form

$$C : y^2 + xy = x^3 - x^2 - 2x - 1, \quad F = \frac{E_4}{j + 3375} = \sum_{n \geq 1} a_n(F) q^n.$$

Here E_4 is the weight four Eisenstein series and $q = e^{2\pi i \tau}$. For an ordinary prime $p > 3$, let $u_p(C)$ be the unit root of $T^2 - a_p(C)T + p$, where $a_p(C) = p + 1 - \#C(\mathbb{F}_p)$. Then Zhang conjectured

$$a_{np^\ell}(F) \equiv u_p(C)^2 a_{np^{\ell-1}}(F) \pmod{p^{3\ell}}, \quad n, \ell \geq 1. \quad (4)$$

This surprising phenomenon links the U_p -action on meromorphic modular forms and Frobenius eigenvalues on cohomology of the elliptic curve. The paper proves and explains these types of congruences.

Theorem 2 ([Bor26]). *Fix $E/\mathbb{Q}$ an elliptic curve. There exist meromorphic modular forms $f_0, \dots, f_k$ of weight $k + 2$ with poles at $j(E)$ such that*

$$a_{mp^\ell}(f_i) \equiv \alpha^i \beta^{k-i} \cdot a_{mp^{\ell-1}}(f_i) \pmod{p^{(k+1)\ell}} \quad i = 0, \dots, k$$

where α, β are the roots of the crystalline Frobenius polynomial on $H_{\mathrm{dR}}^1(E, \mathbb{Q}_p)$.

Method and significance

It is key to consider the *rigid cohomology* on modular curves, the theory of *overconvergent modular forms* and the p -adic geometry of the modular curve. Using an idea of Coleman [Col96] involving Katz-Lubin *canonical subgroup*, we can give an explicit description of the action of Frobenius on the de Rham cohomology group of a modular curve X with a system of coefficients $\mathcal{H}$ in terms of the Hecke endomorphism U_p on space of meromorphic modular forms

$$\mathrm{Frob} \circ H_{\mathrm{dR}}^1(X - D, \mathrm{Sym}^k \mathcal{H}) \longleftrightarrow M_{k+2}^D / \theta^{k+1} M_{-k}^D \circ U_p, V_p. \quad (5)$$

The core of the project is translating the initial congruence into a modular machinery where the study of extension of filtered φ -modules by simple pieces ties the analytic and geometric data. As a byproduct of this proof of the conjecture of Zhang, this work shed some light on a modular and explicit action of Frobenius on cohomology that can be relevant for recent work in the *explicit quadratic Chabauty area* for developing algorithms that do not depend on models of the curves [Ren25].

1.3 Spectral aspects of p -adic Maaß forms

[Ongoing joint work with Jan Vonk]

A spectral dictionary

A theme running through my thesis is that the analytic Laplacian Δ seems to have a counterpart in p -adic geometry. A striking instance of this, going back to suggestions of Blasius and Calegari, is the analogy

$$\left\{ \begin{array}{l} \text{Maaß forms } \mathcal{M}_\infty^{\text{an}} \\ \Delta = -y^2(\partial_x^2 + \partial_y^2) \end{array} \right\} \longleftrightarrow \left\{ \begin{array}{l} \text{Overconvergent forms } \mathcal{M}_p^{\text{an},r} \\ \text{"}\Delta\text{"} = \log |U_p| \end{array} \right\} \quad (6)$$

between the *hyperbolic Laplacian* acting on *Maaß forms* and the *Atkin operator* U_p acting on the space $M_p^0(r)$ of *r -overconvergent modular functions*, under which the Selberg spectral gap corresponds to the smallest positive slope of U_p , Weyl's law to the asymptotics of the slope sequence [Wan98], and the nodal domains (Figure 3) of a Maaß eigenfunction [HR92] to the zero locus of an overconvergent one.

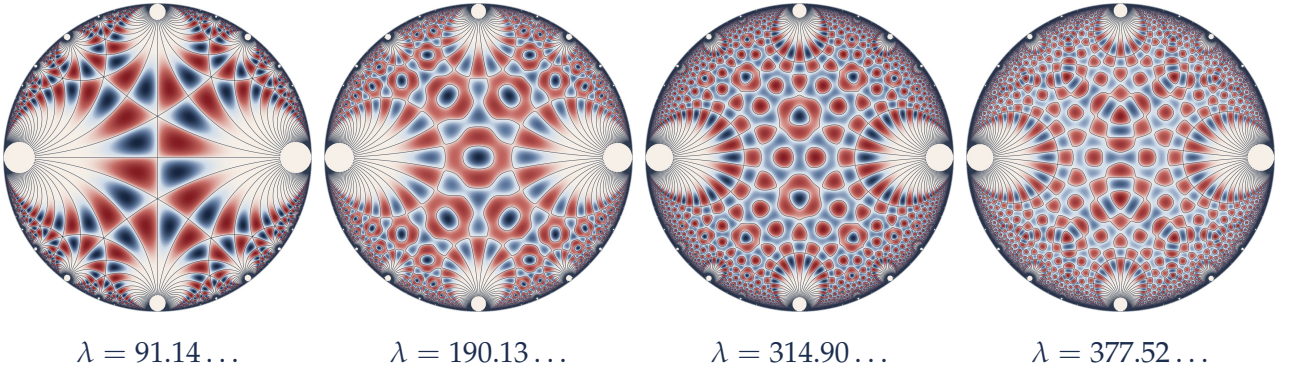


Figure 3: Four Maaß cusp forms for $\text{SL}_2(\mathbb{Z})$, with Laplace eigenvalues λ . The dark lines represent the nodal domains.

Archimedean equidistribution of such zero loci is a theorem for holomorphic forms [Rud05]; on the p -adic side the patterns that have been observed come with no geometric mechanism, and supplying one is the aim of this project.

Exact zero radii

Let p be a prime for which $X_0(p)$ has genus zero, the explicit decompositions of U_p obtained by Buzzard–Calegari [BC05] and Loeffler [Loe07] imply that the eigenfunctions f_n on $M_p^0(r)$, ordered by increasing slope have exactly n zeroes on X_r . We have computed these zero divisors for $p = 2, 3, 5, 7, 13$, and find that their radii not only accumulate at $v(A) = 1/(p+1)$ but appear to obey an exact law: on the component $n \equiv 1 \pmod{p-1}$ we conjecture

$$\boxed{\text{Div}(f_{p^n}) = [\infty] + \sum_{k=0}^{n-1} p^k(p-1) \text{ roots of radius } r_k,} \quad r_k = \frac{1}{p+1} - \frac{1}{\text{lcm}(12, p+1) p^k}. \quad (7)$$

Calegari observed [Cal13] that f_{2^i} converges to the 2-depletion $E_0^{[2]}$ of the weight zero Eisenstein series;

2.2 Zero locus equidistribution for the U_p -operator

Beyond the zero divisor conjecture of §1.3, the dictionary with Maaß forms predicts asymptotic statements: a Weyl law for the slope sequence, and equidistribution of the zero loci as the slope grows, the p -adic counterpart of [Rud05]. My conjecture describes exact radii for one family at small genus; the goal is the corresponding asymptotics, uniformly in the prime and the level, where the geometry of the modular curve at infinite level provides tools with no archimedean analogue.

2.3 Explicit Frobenius and algorithms

The description of the Frobenius action obtained in §1.2 is explicit enough to compute with, and I intend to develop it into algorithms that require no choice of model for the curve, continuing the line of [Ren25]. This is the most computational strand of my work, and the one in which the code I have written for Frobenius on de Rham cohomology and for Gauss–Manin connections in families should be directly useful.

2.4 Long-term synthesis

Taken together, these directions aim to identify the geometric structures that replace the *Laplace operator in p -adic geometry*. The long-term objective is a framework in which U_p -spectra, Frobenius actions and p -adic height pairings appear as different realizations of a common arithmetic spectral theory. Such a framework should connect explicit computations on modular curves with global questions concerning cycles and p -adic L -functions.

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